Biol. Rev. (2006), **81**, pp. 609–628. © 2006 *Cambridge Philosophical Society*
doi:10.1017/S1464793106007135 *Printed in the United Kingdom*
First published online 25 September 2006

On the origin and evolution of major morphological characters

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(*Received* 28 *April* 2004; *revised and accepted* 13 *July* 2006)

ABSTRACT

Although the mathematical principles underpinning population-level evolution are now well studied, the origin and evolution of morphological novelties has received far less attention. Here, a broad but general theory for how this sort of change takes place is outlined, relying on functional continuity, least-constrained components of morphology, redundancy and preadaptation. At least four distinct sorts of redundancy are identified: (i) redundancy arising through duplication (amplification); (ii) redundancy arising through regionalisation (parcellation); (iii) redundancy arising through functional convergence; and (iv) redundancy arising from shared function (functional degeneracy). Although organisms are here recognised to be functionally constrained ("burdened", in Riedl's terminology), these constraints can be overcome through the combination of the four principles given above. Contrary to its common treatment, functional constraint is neither an ever-increasing restriction on the scope of evolution, nor does it require drastic events to overcome or "break" it. Rather, it is an evolutionary quantity, subject to selection at some level. The rules that govern morphological evolution are the primary controls on what is allowed to happen in the evolution of the overall genotype-phenotype system, suggesting strong controls on the sorts of developmental changes that might be associated with macroevolution. This model, then, sees organism functionality as the primary control on evolvability, with exact genetic make-up being of secondary importance. It should prove possible to recast traditional notions of body-plan evolution into the formulations of complex system analysis, which in the future may prove a unifying discipline for fields as disparate as palaeontology and gene regulatory networks. In particular, understanding how morphology can evolve may provide the critical link between the ecological and morphological networks that are currently the intense focus of evolutionary investigations.

Key words: redundancy, preadaptation, constraint, burden, body plan, macroevolution, phenotype-genotype map, *Hox* genes, functional morphology.

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https://doi.org/10.1017/S1464793106007135 Published online by Cambridge University Press

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I. INTRODUCTION

The origins of major body plans (the '*Baupläne*' of Woodger 1945, drawing on a much older tradition) has historically proved a battlefield in evolutionary studies. Orthodox evolutionary theory has held that all organic evolution can be smoothly reduced to changes in allele frequencies through time. Such gradualism has in recent times been challenged on the grounds of tempo by the rise of the 'punctuated equilibrium' theory (Eldredge and Gould 1972); but perhaps the severest test of the orthodox position has been provided by a lack of clear principles about how 'body plan' evolution might occur. Thirty years ago, Gould & Eldredge (1977, p. 147) could write that 'Smooth transitions between *Baupläne* are almost impossible to construct, even in thought experiments; there is certainly no evidence for them in the fossil record (curious mosaics like *Archaeopteryx* do not count)'. It is true that imagining how a frog might be transformed into a prince is testing, but, of course, this is not what we are required to imagine. Rather, considering the phylogenetic history of organisms, one is required to account for what in retrospect appear to be major changes in organisation, only some of which have survived to contribute to the make-up of extant organisms. Without any principles to act as a guide, the feats of imagination required to reconstruct even these intermediates might prove daunting, because, as some of the examples below aim to show, the routes taken in body plan evolution have sometimes been contorted. Nevertheless, it is at least likely that if one could see all the intermediates between two organisms – which would only be possible if the fossil record was complete – the transformation required would no longer strike us as incredible or bizarre: it might even look obvious. Of course, this point needs to be qualified by the problem of how the appropriate lineages would be assembled. As will be implicit in most of what follows, these problems may be solved by a mixture of phylogenetic and functional analysis, the latter falling broadly into what has been termed "reverse engineering" (Stone & Telford, 1999; Csete & Doyle, 2002).

The need for a more general account of how body plan evolution proceeds is all the more pressing because of the rise of another field where such an account is lacking: the evolution of development. Again, as some of the examples below aim to show, the principles that apply to morphological evolution more or less apply to developmental evolution, although there are some special cases of both. It should be noted that with rare exceptions, the principles discussed below are not novel, but have occurred in a scattered, unsystematic way throughout the literature. As such their impact has been greatly reduced, so that virtually none of this material appears in standard textbooks on evolution, despite its potential interest. It is therefore to be hoped that the drawing together of this material will clarify its message and render it more accessible to a greater range of biologists. Given that more than 25 years have passed since the publication in English of Riedl's *Order in Living Organisms* (Riedl 1978), a seminal if idiosyncratic contribution to the field, the time may be ripe for re-evaluation. Finally, it should be noted that major body plan transitions represent an area of evolution that engenders most public scepticism about evolution, and has given most encouragement to creationist objections to evolution as a whole (Gilbert, 2003). The lack of a general theory about such transitions has contributed to the gap in understanding of evolution between palaeontologists and developmental biologists on the one hand and population biologists on the other. I therefore also attempt to show how understanding the principles of morphological evolution might go some way towards bridging the gap between these different disciplines.

(1) Types of evolutionary questions

Evolutionary scientists can broadly be divided into two distinct camps, one occupied by population geneticists and ecologists, and the other by palaeontologists, functional morphologists, and morphological developmentalists (Gilbert, Opitz & Raff, 1996; cf. Fig. 1.1 in Arthur 1997). For the first group, evolution may be considered to be the process of shifting allele frequencies through time. This sort of evolutionary study relies on modelling the dynamics of such change, together with studying the effects of the behaviour and ecology of a particular organism on such gene flow (i.e. the study of selection, drift etc.). For this group, understanding what effect the construction, development or history of a particular organism has on its evolution is often an irrelevance. Conversely, members of the second group are often uninterested in the details of population dynamics or ecology. For them, evolution is seen as the problem of how a complex organism changes through time. The relevant data for this endeavour are comparative anatomy, phylogenetic reconstruction and geological history; understanding how an organism is constructed, and how it develops. From these data, models of the routes evolution has followed and why may be generated. Typical of the sort question that is addressed by the first camp would be 'how does an initially very low frequency-mutation, which is nevertheless of high adaptive value, spread

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throughout a population?' Typical of the second camp would be 'why have organisms not in general developed wheels?' Classical neo-Darwinian theory has been concerned largely with the first of these approaches, and there has consequently been little truly comprehensive and synthetic treatment of the latter. It is a reasonable assumption to make (although not universally shared) that the second field can be reduced in its entirety to the first: so that all evolutionary change *may* be described in terms of population genetics and allele frequency shift [e.g. "all evolution is due to the accumulation of small genetic changes, guided by natural selection, and that transpecific evolution is nothing but an extrapolation and magnification of the events that take place within populations and species" (Mayr, 1963, p. 586; *cf.* Grantham, 2004 and discussion in Love, 2003)]. Even if this assumption is true, though, the second field is still a challenging, underexplored and absorbing field in its own right, although population biologists do not always perceive it to be so.

Of particular interest recently has been the origin of major body plans in evolution: an inquiry that has ranged from investigation of the Cambrian explosion, when many major clades appear in the fossil record for the first time (Conway Morris, 1998; Erwin, 1993; Valentine, 1989; Valentine, 1996; Budd & Jensen, 2000), through to the new field of 'evolution of development' (Raff 1996; Gilbert *et al.* 1996). Various vertebrate innovations such as the origin of fins, limbs and jaws have also been intensely scrutinised (e.g. Ahlberg, 2001). Simply because systematisation of fossils into "stem groups" leading up to the living crown groups can reveal the routes that morphological evolution has actually taken, fossil evidence, properly considered, is always likely to be of importance in this endeavour (Budd, 2001*a*).

(2) Characters and "biological homology"

Any attempt at providing a theory for how morphological characters originate and evolve must first deal with the primary question of what characters are, and how we recognise them (Wagner & Laubichler, 2001). Given that any elucidation of the history of a character must involve reference to the same character in different organisms, the question of character recognition is clearly part of the homology question. As Roth (e.g. 1988) and Wagner (e.g. 1989) pointed out, any concept of homology must contain more than just "descent with modification" from a last common ancestor. A theory of constraint is also required, in order to account for the rather strict morphological limits that apply to particular characters, and to allow us to recognise them as distinct features in themselves. The question of why organisms are divided into different and relatively distinct characters or "modules" is an interesting one, and is discussed further below. The solution of Wagner and others is to suggest that characters are essentially bounded by developmental programmes, and that recognition of a character ultimately rests on this basis – the so-called biological homology concept (Roth, 1988; Wagner, 1989).

While individual characters may indeed often be bounded by their own developmental programme that effectively keeps them within certain limits (at least short-term), I do not accept that this needs to be imported into the character concept *per se*. Whilst the constraint is undoubtedly important, it does not have to be sourced from development (*cf.* Schwenk, 1995; Wagner & Schwenk, 2000). For example, the mere requirement for members of a population to be able to interbreed might impose certain limits, as might functional constraints (see below). Furthermore, it seems unnecessary to understand the development of a character before one can say it *is* a character, which seems to be one implication of the biological homology concept. From at least the time of Aristotle, it has proved possible to talk about characters and their functions without this knowledge [including one celebrated passage Darwin cites in the first paragraph of the On the *origin of species* (Darwin, 1959)]; and this will be the view taken of characters herein. It will quickly become clear from the discussion below that this view of characters is tied, not to their developmental background, but to their functional roles: hands are distinguished fundamentally from arms because they do different things, not because they develop differently.

(3) Simple and compound character sets: viability and origin

Given the above discussion, one would ideally like to be able to distinguish between different sorts of change, those that fall naturally into the realm of population genetics, and those that lie in the provenance of body plan evolution; thus to distinguish between the two sorts of evolution. This distinction is surprisingly difficult to make, and indeed it is locating the subject matter that has made the study of major innovations so elusive to many biologists. Here I try to locate the problem in terms of variation: which variations occur naturally in populations, and can under some circumstances survive to reproduce, and which "variations", although viable, cannot arise naturally? It is important to distinguish between the origin of a character (spontaneously in a population, or over a long period of time) and its viability. For example, the different limbs displayed by living tetrapods are obviously and perhaps by definition viable features that have in many cases persisted over millions of years; but could not as a set have spontaneously arisen within a population, for reasons that will be discussed below. They are not variations on the morphological range of any one preceding population, but instead slowly arose within a set of preceding generations, which were genetically isolated in time and therefore cannot be treated in aggregate as some sort of huge metapopulation. As a result, such characters are merely similar to each other (and, indeed, are homologous) without being variations that arose within a single population (see Olson, 1960; Love, 2003). Thus, they are different in kind, and not just in scale, from population variations, and logically deserve a different explanatory background.

Conversely, some character states can emerge spontaneously in a population but might only be viable under some circumstances. For example, under the selection pressure of reaching high leaves on trees, limb length in herbivores

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might be free to vary between long and short according to the maximum tree height in the environment. However, if all the trees are tall, then having very short limbs will clearly not be viable. Such character states are commonly going to change in a quantitative manner, although not invariably so: eye colour might change qualitatively, for example. Further, this quantitative change is subdivisible into extensive and intensive characters [using the terminology of Kant in the first half of the "system of all principles of pure understanding" section of the *First critique* (Kant, 1781 – see e.g. Bennett, 1966 for discussion)]. In this view characters may maintain their physical extent, but still vary in terms of, for example, their mechanical or physical properties. In other words, characters can grow and shrink, as in hair length, but also fade and fortify, as in colour and cuticle rigidity.

It is tempting, given the above, to think of the characters of an organism as the substrate for character states that are the entities that evolve in populations. If so, then there is an ontological difference between population variations and the homologous characters of comparative anatomy, although the boundary dividing them is indistinct: all characters arose out of population variations. If this analysis is correct, then there should be a science of how these ontologically distinct characters originate and evolve. Nevertheless, this distinction does not quite do justice to the possibility of small-scale but genuinely new features of an organism viably appearing in a population, such as a new bristle on a *Drosophila*.

The various modifications of limb types seen in tetrapods, such as the forelimbs of the horse, mole and human are once more instructive. Although all these modifications might be regarded as different character states of the same feature for the purposes of a cladistic analysis, there is no obvious single selective pressure that could transform one to the other. The reason for this is clear: they are all modifications for particular and differing functions; as shift in function must inevitably lead to shift in selective pressure, the possibility of continuous change under one influence is unlikely. Even if one could perhaps find a set of intermediates between a horse hoof and a mole talon, and show how each transformation could be effected by some single selective pressure such as 'is better for digging', this would not help to explain a horse-human or a lungfish-bat limb transformation. Further, such changes would not be reversible. There could not be a selection pressure that could be defined generally enough to account for changes that can only be described as getting better or worse at a particular function (*e.g.* selection pressure defined by the maximum tree height can account for both increase and decrease of leg length).

Variations that can viably arise in populations might be called simple characters, and those that can only emerge over a period of time might be called compound characters. Are compound characters sets *decomposable* in the sense of Valentine & May (1996) to sets of simple characters (i.e. can they be fully functionally represented by consideration only of their subsets), and how might this be achieved (Laubichler & Wagner, 2001)? This will depend on whether or not selection pressures are themselves decomposable: is the selection pressure acting on a limb, for example, merely the sum of individual selection pressures acting on the constituent bones, muscles, nerves and so on, or is this too subtle a distinction for that particular selective pressure to make? Furthermore, insofar as nerves for example do not owe their current form to a selection pressure such as 'get better at digging', but to much earlier selective events, probably in the Proterozoic, the cause of their form now cannot be the present selection pressure. In other words, considered at this level, the constituent components of a compound character set were generated preadaptively, or are 'exaptations' in the usage of Gould & Vrba (1982). One might wish to consider whether there are any structural or morphological aspects of the members of a character set that might make it compound as opposed to simple, or is the distinction to be posed entirely in terms of reaction to selective regime? For example, one instinctively feels that character states relating to hairs might be more likely to be simple than those relating to limbs, simply because organism functionality is more dependent on limb function than on particular hair function. Sheer construction might have a role to play as well: do concepts such as internal constraints and degree of modularity have a bearing on the compound nature of a character set? These are clearly subjects that need to be addressed in future examinations.

The problem of body plan change, in this view, is to understand the origin and maintenance of members of compound character sets. It should already be clear that the corpus of microevolutionary theory is not sufficient (although normally considered necessary) to explain this type of evolution because it is concentrated on simple character set evolution, *i.e.* selective and other forces acting on populations. So the first important conclusion to draw is that population biology cannot explain change in compound character sets, only in simple ones. To put it another way, the genetics of natural variation in populations ("the problem of variation" of e.g. Stern, 2000) are not directly applicable to the problems of the evolution of compound characters sets except in the uninteresting sense that some subset of all population changes eventually underlies their origin. In other words, major characters are not simply population polymorphisms writ large. A theory of compound character set evolution must elucidate what filters apply to cut out most population changes, and thus set the boundary conditions for macro-evolutionarily relevant micro-evolution.

This definition of a compound character set is related to, but differs from that of Thornhill & Ussery (2000), who discussed characters of irreducible complexity that could not be evolved without certain conditions being met. I have chosen to take an adaptive and variational rather than an engineering approach to the topic in the hope that this will allow the subject to be integrated more readily into mainstream evolutionary thought. Nevertheless, complex characters in either scheme tend to have internal integration that renders understanding how they might evolve problematic. A good example is provided by a human construction, an arch supported by a keystone at the top (Fig. 1). Without the keystone, the arch will collapse, and so the sides could not be erected without it in place. Thus the question of

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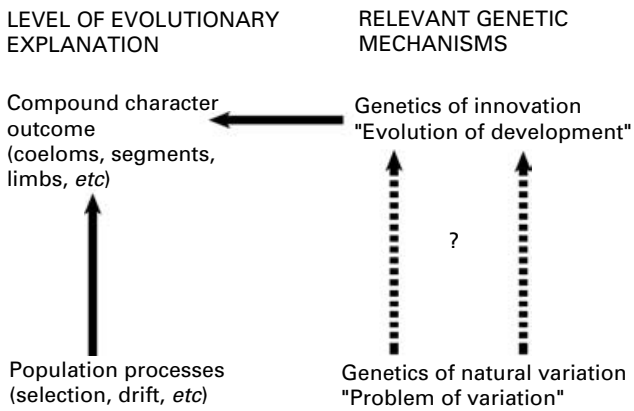

Fig. 1. Levels of explanation in evolutionary biology. Population-level processes such as drift, selection and mutation generate and change the frequency of natural variations. The specific mechanisms behind such variations, however, remain surprisingly problematic (Stern, 2000). A false analogy with this level of change might suggest that there is an equivalent, "higher" level of genetic change associated with macroevolutionary outcomes such a major innovation. However, these changes do not occur in populations, and are likely to consist of filtered sets of lower level changes. The challenge of understanding the genetics of major innovation, then, is to understand the filters, represented by the broken arrow.

"what came first, the keystone or the arch walls" can often appear problematic to the naïve evolutionist.

Given that almost all morphological features are complicated, and that homologous features probably normally form compound sets in the sense given here, and are individually complex as defined by Thornhill and Ussery (2000), this problem is highly pervasive in biology. Riedl (1978) devoted his remarkable *Order in Living Organisms* to this problem (see also the excellent account of Riedl's work by Wagner & Laublichler, 2004). He called the degree of interdependence of characters on each other "burden" and attempted to quantify it using information theory. In his view, burden was particularly a feature of characters that were central, both metaphorically and literally, to the organisation of an organism (Fig. 3). To take a human example, the vertebral column is highly burdened because of its important role in support, flexibility, the nervous and blood vascular systems, and muscle attachment. Conversely, human hair is on the periphery of the organism, and few other systems depend on its precise features. Thus, hair is less burdened than is the vertebral column. It was Riedl's view that burden was a feature that tended to increase through evolutionary time, and was essentially irreversible, so that through time, evolution tended to shift to more peripheral, less-burdened characters. One potential implication of this result, although not one that Riedl himself necessarily drew, is that evolutionary innovation might decrease through time in a particular lineage. This pessimistic view of evolutionary potential has proved attractive to several authors, and implies that in order for major innovation to occur, considerable "resetting" of burden has to take place, perhaps by heterochrony (*e.g.* the hypothetical origin of the bilaterians from a planula-like larval stage of the cnidarians).

Whilst popular, there is little evidence to support this view of evolution. The very examples that are often urged to demonstrate hopeless constraint often also generate counter-examples. For example, even the vertebral column has on occasion been massively modified, such as in the chelonids, or even virtually lost, as seems to be likely for some agnathans. The coelacanth *Latimeria* also offers an interesting example where the expansion of the notochord and its retention into the adult has allowed the reduction of the vertebral column. Molecular development is also replete with such examples, such as the dramatic changes implied in early development of polyembryonic wasps compared to what is known from dipterans (Grbic, Nagy & Strand, 1998; *cf.* Wagner & Misof, 1993). Mere demonstration of these discrepancies however, does not resolve their origins.

(4) Previous views of the origins of compound characters

The problems of the origins of compound characters have long been recognised, but the ways in which they have been treated have not always been satisfactory. In mainstream evolutionary thought, they have been tackled in three broad ways (Levinton, 1988; Budd, 1998) (1) 'key innovation', where one particular innovation enables the rest [see Bond & Opell (1998); Hunter (1998) for a recent case and review; but see also Budd (1998) and Kemp (1985) for criticism]; (2) 'hopeful monster' where a large shift in a complex system is accomplished in one rare leap (Arthur, 1997, Goldschmidt, 1940, Gellon & McGinnis, 1998; see Akam,1998; Budd: 1999; Stern (2000) for criticism); or (3) 'correlated progression', where, subject to central constraints, the entire system can evolve (Kemp, 1982, 1985; Budd, 1998). Of these, only the third really provides a mechanism, in the sense of providing an explanation of what future change is allowed, as opposed to simply describing an evolutionary pattern (key innovation) or allowing that change that does not conform to any principles might occur (hopeful monster). A hopeful monster is perpetuated by a laboratory-based view of developmental evolution that is still widespread, especially amongst developmental biologists (see Stern, 2000). If a particular morphological change is observed during ontogeny, it is likely to be the result of a single mutation. Conversely, the origins of compound features in phylogeny may well be more complex (Budd, 1999; Stern, 2000). An example would be comparing marine mammals that have lost the pelvic girdle and associated limbs, such as manatees, with terrestrial mammals that possess them. If a terrestrial mammal is born without any trace of hindlimbs or pelvic girdle, then the only conclusion in the absence ofan external agency such as poisoning is that the structures were lost as a result of a single event of genetic mutation (see also Stern, 2000). However, the lack of such limbs in manatees cannot simply be explained thus: indeed, knowledge of their phylogenetic position indicates that the limb loss in this clade arose through progressive modification and reduction of the limb structures, with an associated highly complex history of genetic change. Thus evolutionary processes can

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mimic ontogenetic change; only knowledge of their history can actually clarify whether a particular ontogenetic change was responsible for the observed morphological change, despite the constant temptation simply to equate the two (e.g. Cohn & Tickle, 1999). Ontogenetic features, such as heterochrony and homeosis have sometimes been misapplied in this way (for homeosis, see Budd, 1999; for heterochrony see McKinney & McNamara, 1991). In other words, the concept of constraint in the sorts of mutations that are allowed in ontogeny (i.e. are non-lethal) should not be uncritically transferred to concepts of constraint in evolution.

Despite this critique, I do not wish to deny that ontogenetic processes are important in understanding the mechanisms of compound character change. Rather, the aim here is to investigate the ways in which compound characters are allowed to change; this should provide a framework within which that permitted change actually comes about through ontogenetic modification. The rules governing compound character evolution should thus first be considered independently from their genetic background.

Nevertheless, as is increasingly recognised, a satisfactory account for evolution has to encompass both phenotypic and genotypic evolution (e.g. Wagner & Altenberg, 1996), and thus has to move beyond an assumption that the two will simply evolve hand in hand. A radical critique of the entire 'genocentric' view of complex character evolution is provided by the structuralist school who suggest that there is a large or even exclusively non-genetic component in how these characters arise (e.g. Seilacher, 1991; Müller & Newman, 1999, 2003). In this view, structures emerge from complex dynamic processes, and then are stabilised by genetic processes. The difficulty with this view is that, whilst it may hold true for small-scale or general structures (for example, the stable shape of an irregular echinoid test), it seems unlikely that a structure as complex as the vertebrate eye or vertebral column could, even in rudimentary form simply emerge from these processes. These intricate and integrated structures can be termed "designoids" in the sense of Dawkins (1986). Nor, on population genetics grounds, can it be considered that their structure at initial emergence would be rough, and was "fine-tuned" by later selection, once genes became involved in their assembly. The several appeals to events such as the Cambrian explosion, when ecological and selective controls were allegedly much looser (e.g. Arthur, 1997), can be decisively refuted by the increasing recognition that the Cambrian fossil record shows assembly of continuously functional major body plans, not their sudden appearance (Conway Morris & Peel, 1995; Conway Morris, 1998; Budd, 1998; Budd & Jensen, 2000), with recognisable ecological structures already apparent (see Conway Morris, 1986). Further, this model supposes that selective processes are purely destructive, merely weeding out complex structures that have already been generated by non-genetic processes. Part of the aim of this review is to re-emphasise how constructive selection can be, even in generation of true complex novelties (for a recent modelling approach to this problem, see Lenski *et al.*, 2003, although these authors do not discuss higher levels of structural organisation). Nevertheless, the great merit of structuralist approaches is that they break the link between genotype and phenotype, which if rigidly maintained, would make organic evolution in a selective world impossible. Herein, a middle course is taken between a genotypic- and a phenotypic-dominated view of complex character evolution. I first discuss what general controls and principles act on complex characters considered phenotypically, and distinguish several distinct but categorisable types of change. Secondly, I consider what implications these principles have when the genotype and phenotype are considered jointly as a combined system; and lastly discuss how all these factors might relate to the origin and maintenance of evolutionary novelty.

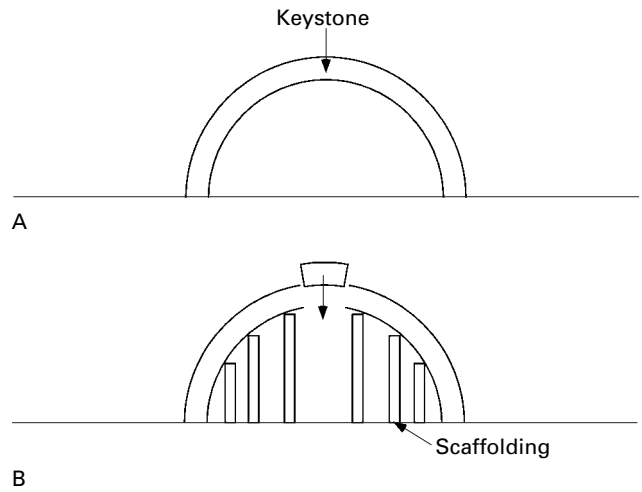

Fig. 2. The arch: a simple example of a complex character in the terms of Thornhill & Ussery (2000). (A) Fully assembled, the arch cannot be simply evolved because the sides of the arch would collapse without the keystone in place. (B) Scaffolding supports the walls until the keystone is in place, at which point it can be removed (elimination of redundancy). For the possibility of placing the keystone on the ground first and inserting wall blocks under it one by one, see Thornhill & Ussery (2000) and Sections II.1.

II. GENERAL PRINCIPLES OF MORPHOLOGICAL EVOLUTION

To return to the example of the arch, the solution of how construct it is simple (e.g. Cairns-Smith, 1985; Thornhill & Ussery, 2000). The existence of the mutually dependent sides and keystone obviously require the previous existence of some sort of scaffolding that allowed construction of the sides without the keystone being present (Fig. 2). Thornhill & Ussery (2000) refer to this mode of evolution as elimination of redundancy, and indeed redundancy plays a critical role in the arguments below. When the sides of the arch are complete and the keystone is in place, there is redundancy in function between the supporting keystone and scaffolding; either one can now be removed without functional failure. This simple scenario exemplifies the widely recognised

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importance of redundancy in the evolution of complex structures, but it is not the only important feature. Indeed, engineering examples tend to have shortcomings when applied to real biological examples because they assume the builder can always produce the required material to complete the structure. If the arch was a biological example, for example, why would the keystone evolve in the first place if not to support the arch (a job it is initally not required to do)?. The model therefore requires expansion, and although the field is not well-studied, four general principles can be extracted that are considered in detail below.

(1) Functional continuity

All evolution at this level – here considered to be the evolution of higher taxa – must conform to biomechanical principles, so that at each stage of evolution a functioning organism must result; this is the principle of functional continuity. Frazzetta (1975) famously considered organic evolution to be like the "continuous improvement of a machine while it is running". This analogy is not strictly accurate, because evolution does not work on single organisms, rather on lineage of descent. Nevertheless, the genetic continuity of information from one generation to the next renders it apposite. It holds true even if 'nonadaptive' stages are passed through (Simpson, 1944). The degree to which functionality is constrained by population considerations is encapsulated by "Fisher's Law" (Fisher, 1930), which implies that large phenotypic changes are exponentially likely to be disastrous for the organism. Some large changes that affect the whole organism, such as heterochronic shifts or dwarfism, may occur, probably because these sorts of changes may not affect the internal relationships of the various parts of the organism, and therefore are not necessarily disruptive (*cf.* Hanken & Wake, 1993).

A lineage can only be viable and persist if each generation is functional – thus, there must be functional continuity from one generation to the next. Such a view is at odds with traditional ways of thinking about body plan evolution in which the inherent constraints of a system are identified, then the directionality of constraint is applied to deduce the evolutionary direction. An excellent example of this is provided by the arthropods, where the strong constraints imposed upon overall body plan by the sclerotized exoskeleton are often assumed to have been evolutionarily causal in other aspects of body organisation (e.g. Willmer, 1990; Brusca, Brusca & Haver, 2003; *cf.* Gudo, 2002). It has thus been claimed that because a septate segmental coelom, external cilia and peripheral muscle system cannot function in conjunction with an exoskeleton, the acquisition of the exoskeleton must have forced the loss of these systems. However, during the period of system change, the functions of any organs that cannot function with an exoskeleton would not be performed, thus the principle of functional continuity would be violated.

A more subtle solution would be if the exoskeleton successively hardened, with the other organ systems also undergoing gradual changes. In the engineering example of the arch, this would be equivalent to building an arch by first placing the keystone on the ground, and then gradually inserting the individual blocks making up the sides of the arch on alternate sides. If the blocks are small, then the functionality of the arch may be maintained (Thornhill & Ussery, 2000). This account is superficially more plausible, but is flawed for two reasons: first it may contravene the principle of asymmetrical dependency (Section II.2), and secondly it ignores the problem of functional thresholds (Jaanusson, 1981; Budd & Jensen, 2000). Broadly speaking, muscles and the exoskeleton form a functional system with particular mechanical regimes to which they are adapted. Although the two components together form a system, they can each be considered separately too. It is not certain that a change in mechanical regime that favours a shift in one system will also favour a shift in the other. Furthermore, because the systems are functionally linked, there would have to be linkage in the evolutionary pathway they both follow during the regime change; and it is not clear that sets steps required to shift from a soft to a hard exoskeleton are appropriate in the shift from circular to lever-style musculature. A small shift in exoskeleton hardness, for example, might simply cause catastrophic failure of circular muscle function. In other words, this style of evolution is either extremely lucky, or it demands too much of the evolving system.

Another example might be system evolution in shifting from very small to very large size in a bilaterian. Consider the linked systems: coelom, metanephridia, blood vascular system and segmentation. These systems relate to each other in a clear and partly definitional way (Fig. 4). In order to have metanephridia and a blood vascular system, a coelom is first required; the metanephridia require a blood vascular system too, with segmentation being dependent on the presence of a coelom (by definition) and the blood vascular system too. A tiny animal (less than 0.1 mm long, say) could have none of these systems, which is why complex metazoans could not have arisen at this body size (Budd & Jensen, 2000; *contra* Boaden, 1989). As the animal hypothetically enlarged, there would be a point where each of these systems would become mechanico-physiologically favoured over a diffusion or ciliary alternative. However, the size at which each separate system becomes viable may not occur in the same order as their functional interdependence demands. As the organism increased in size, a coelom would have to develop first, then a blood vascular system, then metanephridia and segmentation. But suppose the requirements for blood transport demanded that a BVS system was the first to evolve, before the mechanical requirements for a coelom? It is not clear whether the systems as we see them today could have been assembled in this way. Evolving many linked systems in tandem in a smooth evolutionary transition therefore seems problematic. An alternative is provided by the principle of asymmetrical dependency (see below).

(2) Complex system evolution and asymmetrical dependency

The second principle is that for compound components of the system to evolve, there must be flexibility in at least one section of the component and that is the section that will

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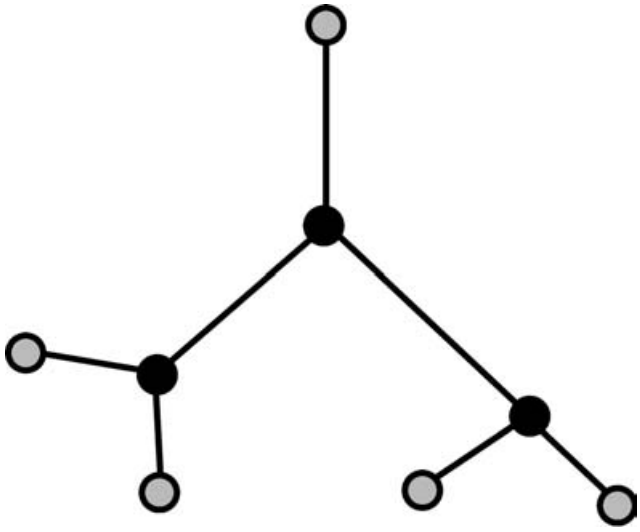

Fig. 3. Burden and position. Character complexes (represented by the circles) can be considered to be more central to overall organism function if they lie at the core of the organism (black circles) than at the periphery (grey circles).

initially evolve; the *principle of asymmetrical dependency*. Whilst it might seem possible that organisms are entirely structurally complex in the sense of Thornhill & Ussery (2000), in practice this is extremely unlikely to be the case. This, as Riedl (1978) recognised, is because multicellular organisms are spatially heterogeneous, with both a core and outer branches of anatomy. There must be peripheral structures; and these structures by virtue of their position are unlikely to be bound up by at least functional constraints (Fig. 3).

The analysis of the organ systems of bilaterians in Section II.1 reveals an interesting aspect of such systems: even in complex functional systems, the integration between the components is typically not a perfectly symmetrical dependency. Fig. 4 shows that the metanephridia and BVS of an organism are functionally integrated, but whereas the metanephridia require the presence of the BVS (or a functional equivalent) the converse is not true. Similarly, whereas a eusegmented organism requires a BVS (to circumvent partitioned compartments), the converse is not true. These dependency asymmetries are reflected in the distribution of such systems throughout the animal kingdom, so that no segmented organisms are known without a BVS (Ruppert & Carle, 1983), and while some organisms with a BVS do not have metanephridia (e.g. the arthropods), the converse is not true (Ruppert & Smith, 1988). Similarly, while the presence of an exoskeleton excludes the operation of circular body wall musculature requiring an internal lever-system of musculature instead (as in the euarthropods), plenty of organisms exist that possess complex internal musculature without an exoskeleton – some annelids and caterpillars, for example (see Budd, 1998 for review). Within a functional system, it is clear that it will be the 'necessary' component that must have evolved first, and the 'unnecessary' component that followed. This implies that the 'necessary' component can operate in two different

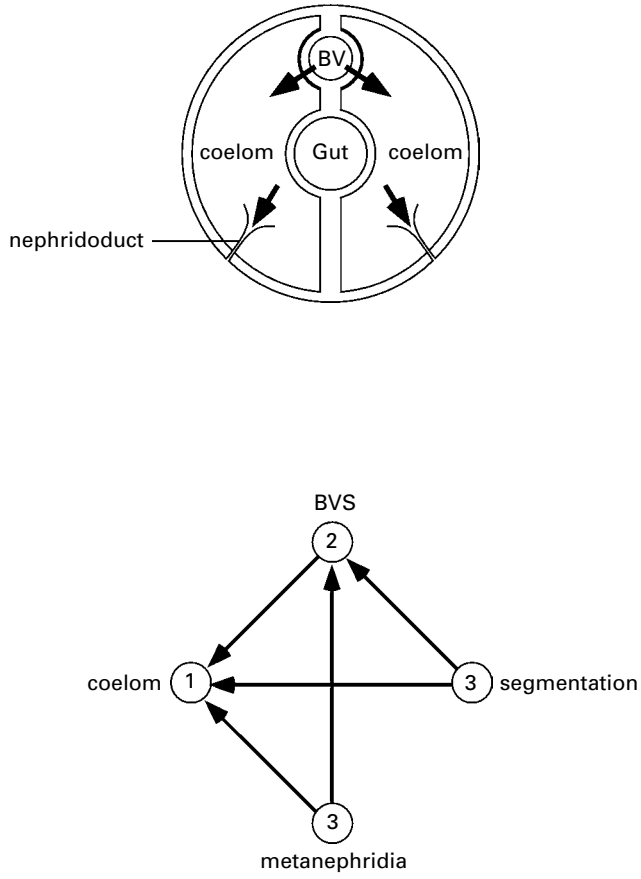

Fig. 4. The organ systems of a segmented coelomate with metanephridia. (A) Transverse section: ultrafiltration of blood under pressure from the blood vessels (BV) occurs into the coelom; excretion occurs through nephridoducts to the outside. (B) The implied functional dependencies between the systems. Fully segmented coelomates must possess a blood vascular system (BVS) so that transport can bypass the compartmentalised coeloms; eusegmentation also implies the presence of a coelom. The metanephridia will involve the coelomic cavity and the BVS. The numbers give the implied order of acquisition of these features, or their functional equivalents.

functional systems. In the terminology of this review, the unnecessary component is the least constrained, and by hypothesis will be the component of the system to evolve first once the system is in place, although the last to be emplaced into the system. In other words, the unnecessary component exercises a functional constraint on the necessary component, whereas the converse is not true. No matter how tightly integrated and interdependent a system might seem, it will still have many points of weakness – some of them very subtle – upon which selective pressures can act to cause change.

It is hypothesised that this is the basis of morphological change by natural selection. It should be noted that population ecology interacts with body plan evolution at this point: ecological niche occupancy will determine which least-constrained component will be placed under selective pressure. Conversely, exerting even strong pressure on

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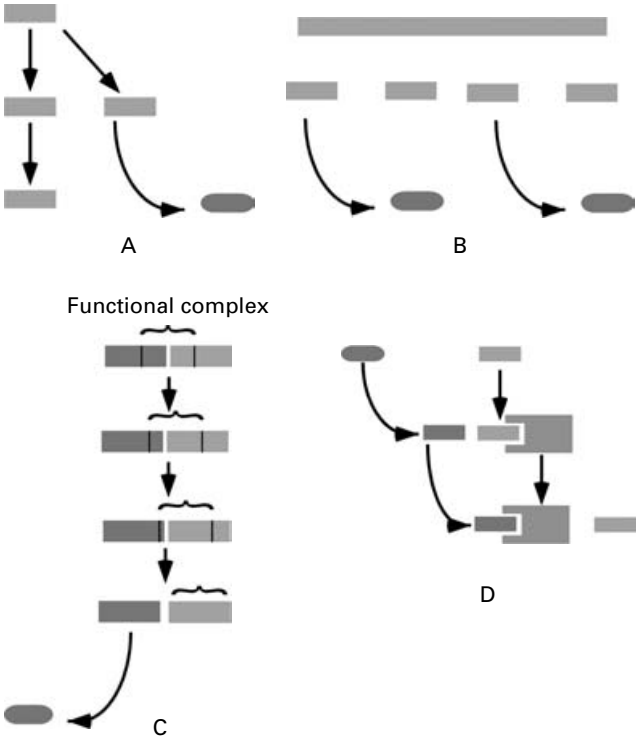

Fig. 5. Four types of redundancy. (A) Amplification, where duplication of a component frees one of the copies for alternative deployment. (B) Parcellation, where an initial structure is modularised, allowing some modules to be redeployed. (C) Functional degeneracy, where the involvement of two or more components in a functional complex allows the smooth transfer of function to only one of the components, freeing the other. (D) Functional take-over: progressive modification of a component allows it to mimic the activitiy of another, and thus to take over its functions, freeing the original component for redeployment.

highly constrained components is unlikely to effect change. Functional asymmetry will thus introduce evolutionary flexibility into a system without requiring redundancy, which is normally assumed to be necessary (see below).

(3) Redundancy

The concept of redundancy, although it is much discussed (Raff 1996; Conrad, 1990; Averof, Davies and Ferrier, 1996), has not been critically examined. In particular, the complexities of the developmental hierarchy in evolution have led many to argue for a prime role for redundancy, seen as gene duplication, for the relief of system constraint. I shall argue below that this is a relatively unimportant mode of constraint relief, although redundancy itself is critical.

The term 'redundancy' has been used variably and without real precision and its importance has thus been obscured. In fact, at least four distinct types of redundancy may be identified (Fig. 5), not all of which might be evolutionarily important. (1) Amplification or simple duplication of a component. The system may thus become overdetermined, so that the original function can still be maintained, while freeing the second component to evolve a new function. (2) Parcellation occurs when a structure carrying out a function becomes modular, with the total number of modules being able to more than adequately carry out the original function: some modules could then be modified for a different purpose. Here, repeated modules emerge not by duplication but by subdivision of an originally larger structure. A good example is provided by the human vertebral column, where the function of flexibility does not require all vertebrae to be free; the lower ones may thus be fused without functional loss. (3) Functional degeneracy, where a function is performed by two or more essential components in combination (Budd, 1998). The burden of performing the function may be able to pass smoothly from one component to others, so that the former is freed. (4) Functional take-over, when the function of a system component is replicated by another component, even though only one of the systems actually carries out the function. Here, redundancy can evolve, and although such evolution does not mean that the two components are initially involved in the same function, either becomes capable of performing it, so that causal function may pass from one to the other without disruption. Budd, 1999 gives a theoretical example in homeotic mutations of how this situation might arise: far from being unusual, the broad correspondence between morphology and its generative developmental hierarchy (see below) suggests that it might occur surprisingly often. The origin of the mammalian middle ear, for example, seems to involve the evolution of a second functional craniomandibular joint (between the dentary and squamosal bones, as well as the original quadrate-articular one) as well as progressive functional shift through functional degeneracy; allowing joint function to be transferred from the old to the new joints (Fig. 6; see discussion in Rowe 1996 and Thornhill & Ussery, 2000).

Of these four types of redundancy, amplification is usually considered to be at the heart of evolutionary change. However, it does not in itself offer a solution to how the extra developmental or functional capacity can be exploited to produce novelty. Also, merely duplicating components of a system without any selective advantage will presumably be disadvantageous in the short-term for an organism; unless a new function arises the unused component will decay by mutation (see Section III). In morphological systems, such redundancy is common, but is typically a continuous rather than a discrete character, and can gradually evolve (as in parcellation). For example, in an arthropod, trunk flexibility is due to the articulations between tergites. A few of these articulations can fuse and still allow flexibility to be achieved: hence some groups of segments can be used to produce tagmata. This redundancy did not appear as the result of serial repetition of a functional unit (i.e. a segment); rather, segmentation and the articulations between tergites themselves evolved, taking over the function of flexilibity from a pre-existing organ system (i.e. by a process of parcellation rather than amplification; see Wagner & Altenberg, 1996; Albert, Lannoo & Yuri, 1998; Budd, 2001*b*).

Functional degeneracy, on the other hand, seems a more promising route for excess capacity in functional morphology and development to evolve. One example of

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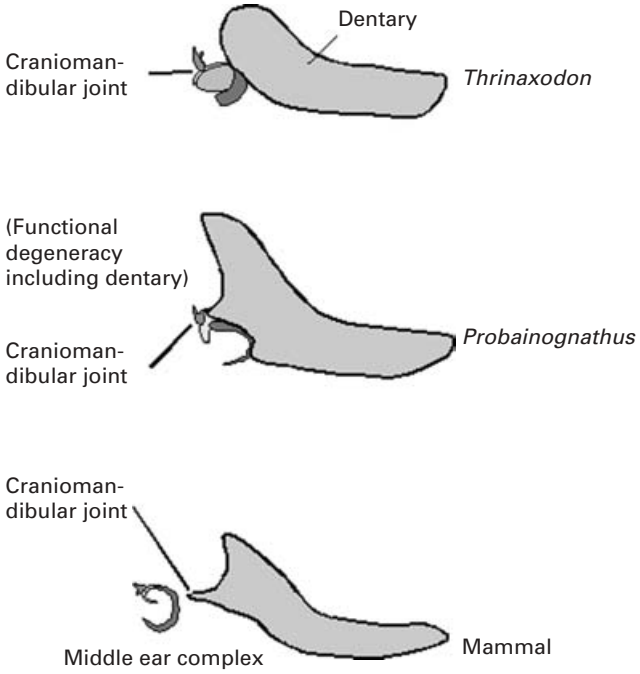

Fig. 6. Functional degeneracy and functional take-over exemplified by the middle ear evolution of mammals (after Rowe, 1996). *Thrinaxodon* and *Probainognathus* are members of the mammal stem group. Backward growth of the dentary involves it with the craniomandibular joint, generating functional degeneracy; function can then be transferred smoothly to the dentary. In versions of this transition where a secondary joint is formed, the process would also demonstrate functional take over (see Fig. 5).

morphological functional degeneracy is provided by the postulated changeover in functional systems between an onychophoran-like hydrostatic haemocoel acted on by peripheral musculature, and the lever-style system seen in euarthropods. From the fossil record, it is postulated that internal musculature evolved for support, but then became capable of sharing the function of movement. Functionality of the entire system could then be smoothly transferred without discontinuity, and the peripheral muscle could be reduced, allowing the eventual development of external scleritisation (Budd, 1998). Such degeneracy is another aspect of the evolution of the jaw bones of mammal-like reptiles introduced above (Fig. 6). This function, being shared between several bones, could be smoothly transferred towards only one of them, the others being released for incorporation into the middle ear. The evolution of cichlid jaws (Galis, 2001) also falls into this category; the dual involvement of epibranchial 4 and the upper pharangeal jaw in biting allowed function to pass smoothly to the upper pharangeal jaw, freeing epibranchial 4 and allowing a much more flexible jaw apparatus. Finally, this sort of mechanism has been postulated for the preservation of duplicated pairs of genes (Force *et al.*, 1999) where each of the pair can undergo complimentary degenerative mutations whilst retaining functionality as a pair (*cf.* Ganfornina & Sanchez, 1999). Among its other consequences, this mechanism may act as a means of storing excess genetic capacity without releasing non-functional genes into the genome, which would be subject to decay from lack of selective maintanence.

Redundancies of the types discussed above may evolve smoothly as well as appear suddenly by duplication. How these redundancies evolve will be subject to the principles of functional continuity and least constraint: a system component will, under the influence of ecological selective pressure (or drift) evolve to potentially or actually share or replicate a function with another system component. This will potentially free the latter component to evolve a new function. Actual redundancy implies that function is actually shared between two or more components, whereas potential redundancy implies that a structure could share a function if the right circumstances come about. Potential redundancy is not an absolute quality; a pre-existing structure can be more or less capable of potentially sharing a function. Potential redundancy arises in preadaptive circumstances, which are considered below.

(4) Preadaptation

This well-known principle applies to characters that can develop a new function based on their previous functionality; a typical example is provided by feathers, which may have evolved primarily for insulation, display or support (Sumida & Brochu, 2000) but were then later co-opted as flight devices. In common with some other authors (e.g. Skelton, 1990) I do not see the need to replace the unambiguous term 'preadaptation' with the term 'exaptation' (Gould & Vrba, 1982), and thus retain the older term, while recognising that it has no teleological implications. In the present context, it is important to stress that only systems in the right place at the right time are able to change function: such components must also be partly or fully potentially redundant compared to the new function they will share or take over, when a completely new function is not evolving (as in the feather case). Conversely, potential redundancy is of no use if the redundant component is not preadaptively sited. This view of preadaptation implies that some structures have the potential for multi-functionality, and these different functions can be exploited at different times (in functional shifts) or simultaneously (in redundant situations).

A good example to demonstrate the interplay of redundancy and preadaptation is again provided by the evolution of the mammalian middle ear (Fig. 6). The bones that were eventually released into the middle ear from the jaw were preadaptively positioned in that they were situated at the back of the jaw; they were also potentially redundant with hearing function because they were small and thus able to respond to vibration accurately. When released from the functionally degenerate system, these bones were able to pass smoothly into a hearing system both because they were near the ear, and because they were already able partly to fulfill a hearing function. Clearly, in order for such evolution to take place, the evolving components must be least-constrained in the sense discussed above, *i.e.* they must have little functional dependency upon them (have low burden). If so, then their evolution can cause redundancy itself to

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evolve, at which point the component of the system with which they are evolving a new interaction will establish a functional dependency upon them. In doing so, a constraint will potentially be lifted from another component of the system. Evolution of least-constrained components will thus generate first redundancies, and as redundancy is eliminated, transfer least constraint to different components of the system (Fig. 7).

One final, and well-known aspect to preadaptation is that it provides the key to understanding how compound characters can be sustained through the long periods that may be required to bring them to their present functional state. For many systems, the answer to the question of "what use is 10% of [the organ]?" (Dawkins, 1986) will be "as a fully functioning component of a functional complex different to the one it now resides in", and not "10% is better than 0%". To give two examples: the forelimbs of tetrapods did not arise to assist stranded fish to struggle from one drying pond to another, but were present (along with most other tetrapod innovations) in aquatic animals (e.g. Coates, 2001); the musculature of arthropods was well adapted not for articulating tergites (which evolved late in arthropod history), but to brace a large body (Budd, 1998). Simulations such as Dawkin's famous "methinks it is like a weasel", where phrases are constantly selected for their likeness to Shakespeare's, are potentially misleading, as Dawkins himself comments (Dawkins, 1986), because the evolution of compound characters, in the sense given here, will always involve shift of adaptive function at some point. To put it another way: evolution is short-sighted and cannot set the long-term goals required for change of compound characters, although they can still be attained as a by-product of short-term gains.

(5) Discussion: the organism as a system

In the above analysis, any evolution in a complex system is suggested to commence with minor variation in the least-constrained component, which will typically continue to evolve until it is no longer the least constrained. With care, a certain degree of both retrodiction and prediction of morphological evolution should be possible from functional analysis (see Stone & Telford, 1999 for a slightly different perspective). Least-constrained component evolution will occur in one of two ways: either the system is locked (under that particular ecological regime and selective pressure), so that this minor evolution does not enable other change; or some sort of functional degeneracy or redundancy will evolve in a preadaptive context. At this point, the function of some component will be freed; and so on until, in theory, the most deeply functionally embedded structures could be released from their constraints (Fig. 7). Constraint is thus an evolutionary concept, and although laboratory manipulation typically cannot circumvent deeply embedded constraints such as the presence of a vertebral column in vertebrates, a long chain of the simple changes outlined above could do so.

As new functional complexes form in this way, some of them very subtle, asymmetries will appear in the response of characters to selection: initially simple characters may start to behave in a compound manner relative to selection pressure, as defined above. In other words, a character may evolve towards one end of its simple range, at which point it may become embedded in a new functional complex, and thus resist any reversed selective pressure that would normally drive it back to its original state, as the new functional dependencies established on it remove its 'least

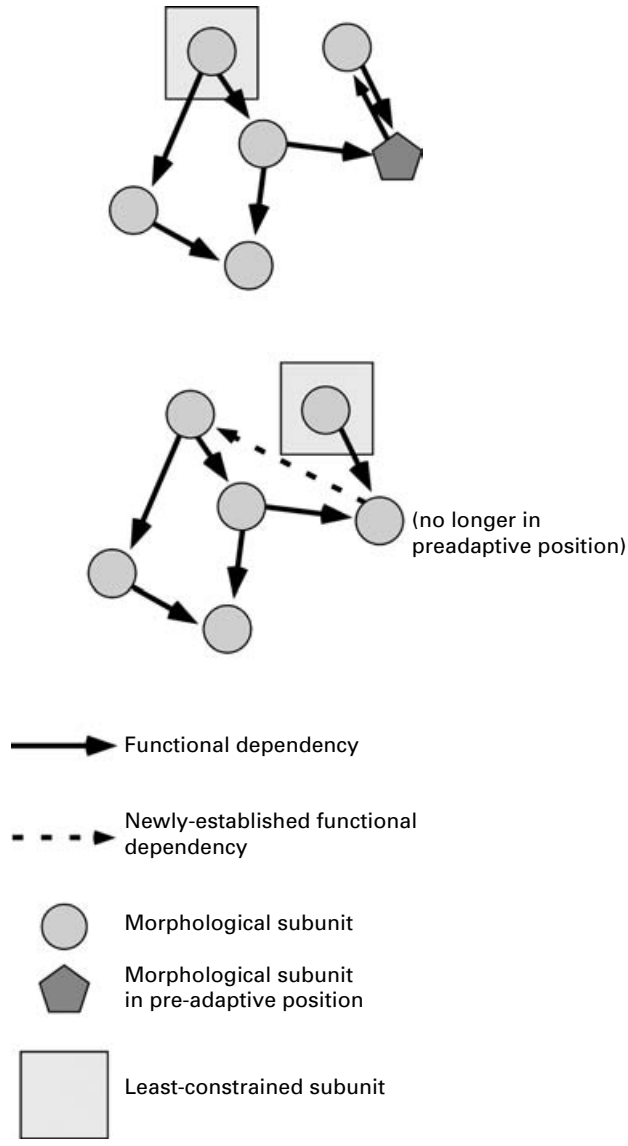

Fig. 7. Least constraint, preadadaption and functional degeneracy/take-over in an integrated system. In the upper diagram, an integrated system possesses a component with no major functional dependencies, making it the least-constrained of the system. As it evolves (lower diagram), another component in a preadaptive position can form a functional dependency on it, causing redundancy with the function it carries out with another component. This former function can then be lost, changing the functional relationships between the components and generating a new least-constrained component. Note the direction of the new functional dependency is always towards the evolving component.

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constrained' status. Such asymmetries may in theory be detectable even in at the population level. This, then, would be one example of how a 'ratchet'-like process could, even during very small-scale changes, capture population variations and render them irreversible in the short term (*cf.* Stearns, 1994). Evolutionary novelties may in this manner be protected against simple reversal, and thus loss, by this sort of asymmetry in the evolutionary process. Such asymmetry may provide an important basis for the development of further novelty out of the smallest scale changes.

(6) Modules and protocols revisited: complex systems analysis of functional morphology

The above view of organisms, then, sees their morphological evolution as essentially changes in interactions between different functional parts. In essence, such a view is compatible with seeing organisms in terms of complex systems, a field that has generated a great deal of literature in recent years. In that terminology, characters become "modules"; i.e. components of the system (following Csete & Doyle, 2002): (1) with identifiable interfaces with other modules called protocols; (2) with the ability to be somewhat independently modified; (3) that can be abstractly modelled; and (4) that derive additional identity from the rest of the system. However, the other feature of modules, that they also retain some degree of identifiability when rearranged or isolated, seems less applicable to morphology; evolution typically does not 'rearrange' or 'isolate' features. The exception would be if homeotic mutations (see Section III.2) were to play an important role in morphological evolution. In the light of the analysis above, it seems clear that one of the principal important features of morphology is not just the modules or characters themselves, but their functional interactions, which can be considered to be protocol-directed. In other words, the identification of features such as asymmetrical dependencies suggests that the interactions between modules are not simple "connections", but have a considerable degree of texture; they have both strength, and directionality.

III. WHOLE-ORGANISM EVOLUTION: THE GENOTYPE-PHENOTYPE RELATIONSHIP

The above model leads to consideration of how organisms evolve as a whole, with particular regard to the relationship between phenotype and developmental genotype.

The relationship between morphology, development and evolution is a subtle one that will be partially explored below. The principal questions that need to be addressed are: how are morphology and the genome structured, and what is the relationship between these two structures? How does the environment interact with these structures? How does the structure of the genome and morphology together control the paths that future evolution may take? Such questions have recently been the subject of considerable interest (see e.g. West-Eberhard, 2003), and cannot possibly be fully discussed here. However, the question of how the morphogical evolution outlined herein relates to the genome is of direct relevance, and this is addressed further below.

(1) Physical and functional structure

Both the genome and morphology are structured (Figs 8, 9), i.e. they have a non-random arrangement of their components. This structuring, however, is of two sorts: physical structure (e.g. *Hox* clusters and the sequence of bones in the tetrapod leg); and functional structure (e.g. the gap gene – segmentation gene – *Hox* gene cascade and the action of muscle on skeletal elements). Physical structure may be discovered simply by producing a physical map of the genome or of the organism as a whole; and functional structure by detailed functional analysis, again of either the genome or morphology. The two sorts of structure are typically not independent, so that, for example, *cis*-operating gene cascades may show both functional and structural connections, and morphology that functions in a complex also typically has a spatial association with the other members of the complex. Neither of these associations are absolute though; so there are *trans*-operating genetic functional units, and functional units in morphology that are not spatially associated (for example, in the mating display plumage of birds). Nevertheless, the function a particular component of an organism has is defined partly by its own internal organisation, and partly by its spatial placement relative to the rest of the morphology. Functional analysis may itself reveal a complex and hierarchical structure, such as the functional complex that is a muscle acting on a limb bone (another functional complex). Functional hierarchies may, however, be decomposable (Valentine & May, 1996) or largely so (see e.g. McShea & Venit, 2001 for discussion).

Morphology has spatial and functional structure no matter how it is generated. However, given that all morphology is generated by an interaction between the environment and the genome (whose respective roles vary greatly), a further feature is present that is generally called correlation: the functional and physical structure of the genome is reflected in morphology because morphology is not free to vary completely independently from the genome. Riedl (1978) referred to this relationship as the "imitatory epigenotype": the structure of the phenotype is mirrored to some extent in that of the regulatory genotype. For example, modular structures tend to have their own package of development bounding them (as in the 'biological homology concept'). The idea behind this view is that evolvability of phenotype is facilitated by a gearing of regulatory genotype to it (e.g. Wagner & Altenberg, 1996. However, this is not the only possible explanation for the correlation. The alternative view suggested here is that developmental systems must deliver certain goods, such as a particular larva or adult morphology, and how well they do this is the key factor in deciding whether or not a particular system is selected. In this view, the efficiency (or some other metric, as yet undefined) of the genome in producing the morphological package required is the over-riding factor, and its structure is therefore determined by the structure of the phenotype – *i.e.*, something that is typically modular. Such a view certainly cannot be accepted without reservation;

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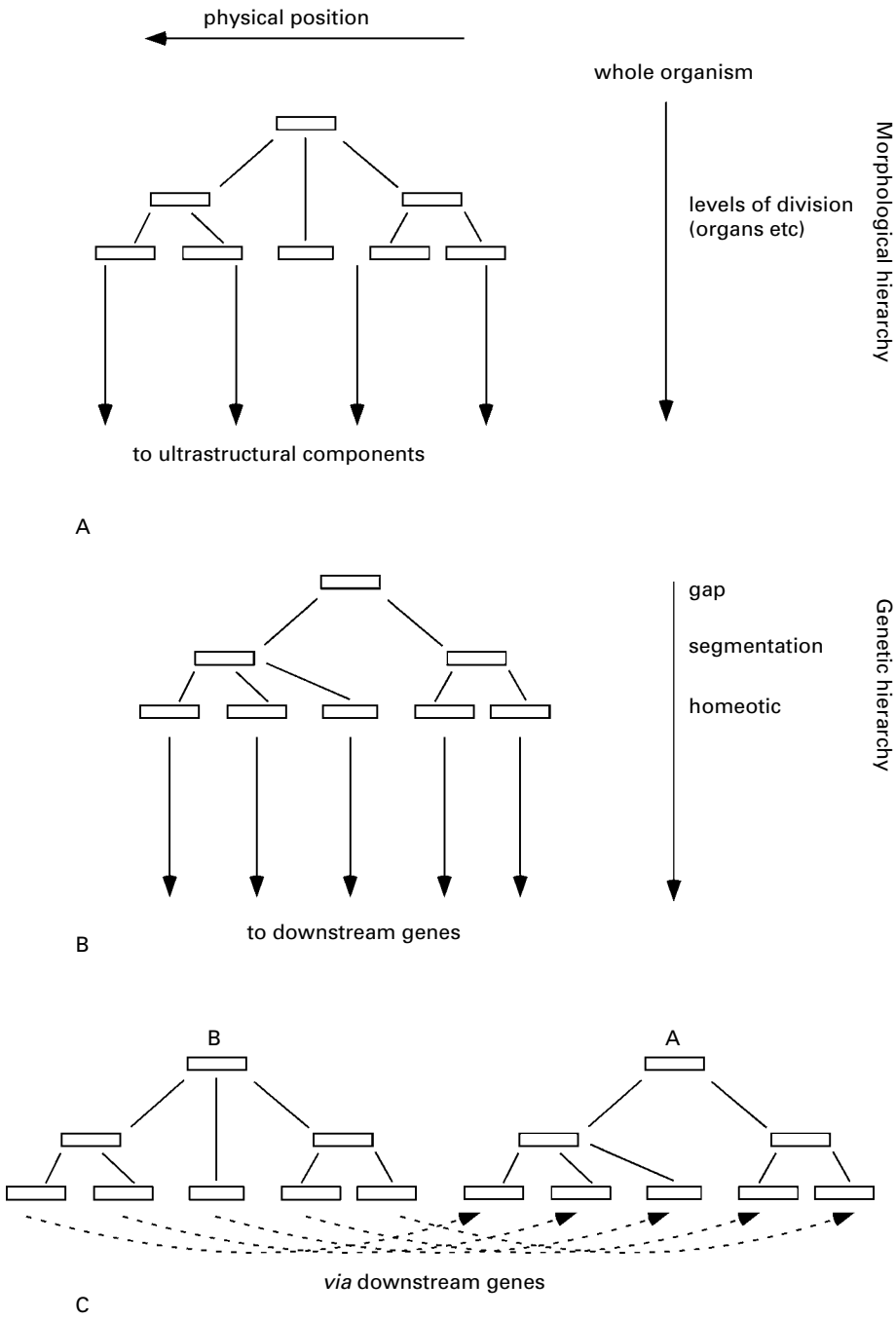

Fig. 8. Morphological and genetic hierarchies. (A) Division of morphology is based on physical position. (B) The developmental hierarchy is based on a cascade of effects from general to more specific. (C) The relationship between A and B is clarified by considering developmental genetics as a physical positioning framework.

given the complexity of the genome, and the amount, for example, of "junk" DNA it is clear that there has been no relentless paring down of the genome to an efficient production of morphology with no other considerations. Further, one might question what exactly is the selective advantage of the 'gearing' of the genome to the modular phenotype. The answer to the latter may be in terms of reliability of production of final product [see Waddington's genetic assimilation experiments (Waddington, 1961), which showed that certain types of genetic structure did produce a particular morphology more reliably and stably] and minimum energy use, but clearly this is an area requiring further investigation by modelling and other approaches.

(2) The problem of variation

Whilst population biology has classically treated the genome as a "black-box" generator of random, small phenotypic

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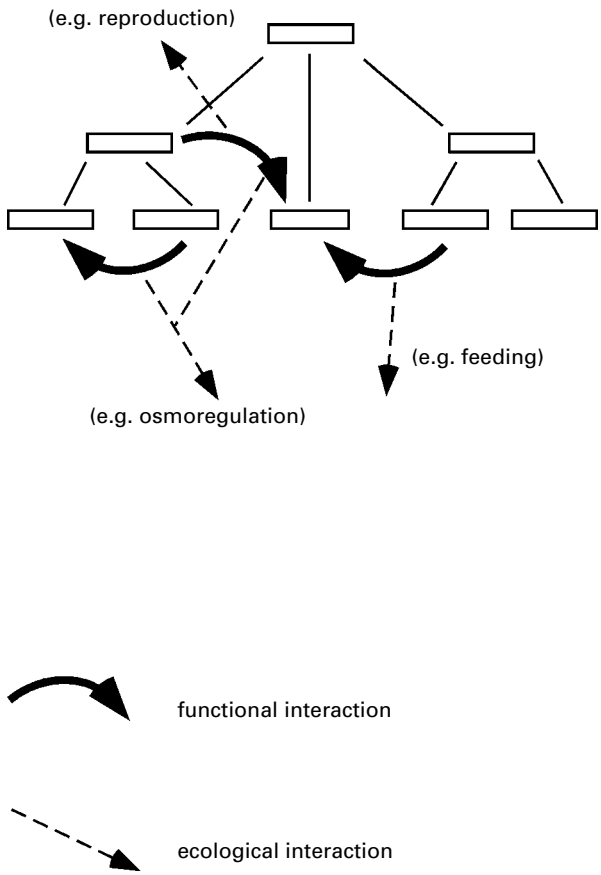

Fig. 9. Interactions between components of a morphological hierarchy to fulfill various functional requirements. The functional requirements act as conduits of selective pressure onto the morphological components. Only if this pressure acts on least-constrained components (see Fig. 7) will evolutionary change occur at the body-plan level.

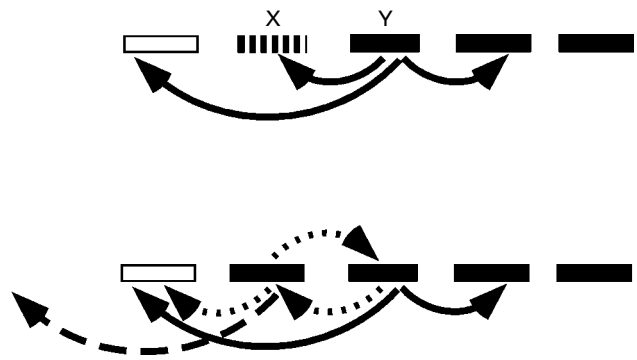

Fig. 10. The effects of homeotic transformation on functional integration. In the upper diagram, five characters are depicted, together with the functional interactions of one of them (y). If character x is transformed homeotically into y, as in the lower diagram, a new and completely inappropriate set of functional relationships emerges, including attempted interactions between the wrong units (dotted arrows) and impossible interactions (dashed arrow), as well as the normal interaction (black arrows). Assessment of such diagrams may allow quantification of the disruptive effects of homeotic transformation.

change, increased understanding of the structure of the genome is beginning to lead to a much more complex and subtle view. Further, the very existence of hierarchy in development suggests that change at different levels in the hierarchy will have differing phenotypic effects. How can we determine which of these effects is evolutionarily important?

As Goldschmidt (1940) showed, the relationship between developmental genetics and phenotype is not a rigid one. Although under particular circumstances a genome will produce a particular morphology (making allowance for pleiotropic and plasticity effects), the opposite is not true. This implies the possibility of phenocopies: effectively, the same morphology mapping onto different genomes. In this context Budd (1999) discussed the Bithoracic morphology in *Drosophila melanogaster* originally taken up by Waddington (1956): even if one sort of genetic change (homeotic mutation) is ruled out by Fisher's Law, it is possible that it could be mimicked by another sort of change, generally a progressive shift involving downstream genes. Under precisely controlled environmental conditions, a particular genotype may generate a particular phenotype, but a particular phenotype can be generated by a variety of different genotypes, with the most obvious example of this being provided by redundancy in the genetic code. Nevertheless, mutations do not necessarily have to lead to phenotypic change, because there is at least some evidence that genuine redundancy may be stably maintained in the genome (e.g. Nowak *et al.*, 1997). If so, then mutations affecting one of the redundant pathways will not disrupt the function of one of the others, and the phenotype might be unaffected. Nevertheless, it is questionable if even in these cases, mutations would have no effect on function and thus potential selective significance. Developmental canalisation might also reduce phenotypic variation to zero even when considerable polymorphisms are present in a population; although this seems unlikely (Wagner, Booth & Bagheri-Chaichian, 1997; Gibson, Wemple & van Helden, 1999).

Other important aspects of the two components are that phenotype is generated by the genotype, and that selection acts on the genotype through the phenotype. These relationships are sources of asymmetry between the genotype and phenotype. A particularly important relationship to stress is that between the functional morphological map of the organism, and the developmental hierarchy. The latter can be considered as a positional map that directs downstream genes to particular physical positions in morphology. This mapping allows the direct functional assessment of changes in the developmental genotype (Fig. 10), and from this is it easy to see how the functional map of the organism might be disrupted even by changes that cause apparently viable morphological changes when considered in isolation. For example, the wings of bithoracic flies seem to be well-formed structures, but in terms of their interactions with the rest of the organism (muscles, nerves, aerodynamics etc.) they are a functional disaster. Such analyses show very clearly that the mere internal functionality of the homeotically transformed structure is by no means the only

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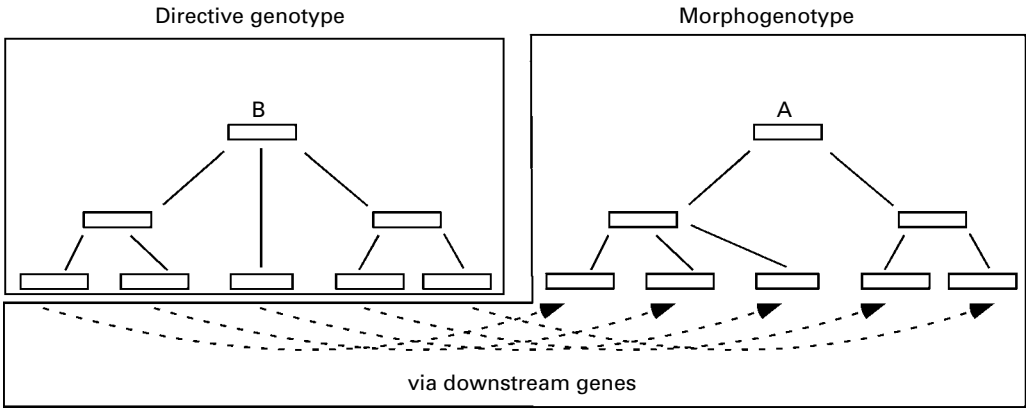

Fig. 11. Division of Fig. 8c into the directive genotype and the morphogenotype (see text for further details). As discussed in the text, the morphogenotype is the least-constrained component of the system, and is functionally dependent on the directive genotype.

factor in determining the viability of the change, but also the implications for functional integration.

(3) Evolution in the phenotype-genotype complex

In a Neodarwinian framework, concepts such as fitness should be used in assessing whether or not particular evolutionary changes are allowed; the magnitude of a phenotypic change is in some way inversely proportional to its likelihood of being favoured. The Neodarwinian synthesis thus places strong restrictions on how these sorts of changes might take place, and so on what sorts of changes might take place. In particular all change, whether in the genome or in the phenotype, can only be selected for through its effect on phenotype: the selective agent – the physical and biological environment – can only act on the concrete organism and its ability to pass on its genetic material to the next generation. Even changes in genome organisation, or in subtly differing interactions between genes and molecules must, in this view, occur only because of some favourable or at least neutral effect they produce in the phenotype. Further, given the lack of an essential link between a particular phenotype and its generative genotype, it follows that analysis of allowed change in the phenotype need make no reference to the genotype. Evolutionary functional morphology is thus essentially autonomous from genome function.

These points suggest that, in terms of determining types of evolutionary change, the whole genome-morphology complex needs to be considered in three compartments: the high-level developmental genotype, incapable of normally generating natural viable variation; the low-level developmental genotype, which is capable of doing so (Fisher's Law or some variant being the arbiter of the division); and morphology itself. Some genes may be classified within both of the genotype compartments, because of their varying roles in development. Hence, *Hox* genes, which classically generate large-scale, non-viable homeosis, should in this role be considered as being in the high level developmental genotype; but their additional role in much smaller scale patterning (Stern, 1998) implies that they can also be considered part of the low-level developmental genotype too. Because of the extremely subtle relationship between these lower levels and morphology, the low-level developmental genotype could be grouped with morphology: the high levels may be called the directive genotype, and the low-level/morphology complex the morphogenotype (Fig. 11).

This grouping allows the whole genome/morphology complex to be reconsidered because it becomes clear that there is a functional asymmetry between the directive genotype and the morphogenotype. Because allowed change can normally be generated from within the morphogenotype (either genetically or, in a broad sense, epigenetically) it follows that the morphogenotype is not constrained by the directive genotype, whereas the converse is not generally true. As a result, the morphogenotype is the least-constrained component of the system. This view is consistent with the idea that morphology is generated by the directive genotype and is therefore functionally dependent on it; changes in the directive genotype definitionally cause functional disaster, but the converse is not true, because of the inherent and incremental flexibility in the morphogenotype. The genotype/morphology complex can therefore also be treated in terms of functional dependency. Again, because this is a functional analysis, it should be stressed that the roles of the relevant genes are important, and not their formal definition.

Simply because of the functional dependency of the directive genotype on the morphogenotype, the rules that determine whether a change is evolutionarily possible must first consult morphology, before asking if the change is allowed on pure genetical terms, even if more purely genetic constraints such as those arising from pleiotropy have importance. If so, then under a particular selective regime, one would expect the morphology to come under pressure first, and its evolution to continue until it was no longer the least-constrained component in the system. This would typically occur because redundancy had appeared in the system, for example, by duplication of the directive mechanism (Budd, 1999). For example, a directive mechanism within the morphogenotype might evolve to be similar to a preexisting one in the directive genotype. It would be at this point that selective pressure on the genotype structure would act to take advantage of this lessening of constraint, and

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allow transfer of the directive mechanism from out of the morphogenotype into the directive genotype.

This model of evolutionary change is profoundly at odds with the fashionable view that change is enabled by simply accumulating changes in the genotype (e.g. Davidson, Peterson & Cameron, 1994; Cohn & Tickle, 1999; Holland, 1998), until a sufficient quantity has accumulated to allow morphological innovation to occur. The obvious problem with such a model is that, if selection is taking place *via* the phenotype, then complexity in the genotype must somehow reflect complexity in the phenotype, but not necessarily *vice versa*.

Rather than morphogenotypic change being enabled by increases in directive genotypic complexity, the converse must be true. Even in cases where a correlation has been claimed between increases in genotypic and morphological complexity, the direction of causality might thus be opposite to that commonly supposed. For example, entire genome duplications have been postulated to lie behind major events in chordate evolution such as the origin of paired fins. If paired fins evolved gradually and adaptively though, it no longer seems necessary for the genome to have increased in complexity, as long as sufficient downstream variation exists to allow microevolutionary change (for a similar argument see von Dassow & Munro, 1999). On the other hand, such phenotypic change would eventually enable directive genotypic change, including perhaps duplication of all the genes. Indeed, such a pattern has now been suggested explicitly for vertebrate evolution (Donoghue & Purnell, 2005).

(4) Genotype-phenotype correlation and the origin of modularity

In a (hypothetical) organism where the components of the system are functionally entirely separate from each other, each may evolve without the constraint of the other components. Such a state may be described as being highly modular. Cladistic analysis has sometimes been criticised on the grounds that it treats organisms in this way, treating the characters of organisms as 'colored balls in a teacup, incidentally associated in some particular combination' (Raff, 1996, p. 136). Extreme arguments against the 'body plan concept' may also be seen in this way, with characters that persist in a particular lineage seen as being merely statistical flukes, rather than being maintained by the constant reliance on them of other features of the body plan (e.g. see Williams, 1992). Conversely, systems also exist where there is a high degree of integration or interaction between different components, and these may be referred to as being highly interconnected. A good example of such a system is the 'phylotypic stage' postulated to typify the development of each of the phyla (Minelli & Schram, 1994; Slack, Holland & Graham, 1993). Such systems are often regarded as being less susceptible to change than are modular ones, because in the former, a wider range of changes may occur in any particular component which do not have a lethal effect (as each component has only few connections with the rest of the organism). Interconnectivity is thus often held to provide one form of constraint, with the implication that interconnected systems are 'less able' to evolve. Highly interconnected states such as the phylotypic stage, are held to be highly conserved throughout many organisms, whereas modular states (such as early and late development) are thought to evolve more easily.

One of the interesting aspects of organismal structure is that despite the lack of a logical connection between genotypic and morphological structure, such correlations in fact generally exist (von Dassow & Munro, 1999). The developmental genotype typically produces a body plan in its broad scale, then gradually adds details. The units of division of higher levels often correspond to real morphological divisions, such as the gap genes and *Hox* gene boundaries coinciding with segmental or tagmatic boundaries in arthropods (notwithstanding the parasegment controversy, see Lawrence, 1988). It would of course be interesting to know how this modularity and correlation arises. From the above analysis, one would predict that discrete morphological structures or modules would tend to evolve first, and directive or downstream genotypic correlation with them would evolve later, or at least lag behind in construction. Therefore, the origin of modularity must be looked for in functional morphology. Morphological specialisation and modularity would appear to be an inevitable consequence of response to selective environment, in that division of labour between different organ systems will allow more efficient performance of different ecological tasks. Even though organisms are integrated wholes then, their functional morphology should evolve towards specialisation. This trend can be seen in many different ways. For example, in sponges and cnidarians, a single cavity and flow of water fulfills an array of functions including feeding, respiration, excretion, support and reproduction. In bilaterians however, these functions become successively compartmentalised (Ruppert & Barnes, 1993) to different flow systems and body cavities, so that in an organism such an annelid, respiration, reproduction, support and feeding are carried out independently, and, presumably, more efficiently. Similarly, the evolution of the coelom allowed separation of function of muscle systems such as the body wall (movement) and the gut (digestion). In vertebrates the process has continued with the isolation of the heart and the evolution of the diaphragm.

Therefore, ecological pressures lead to morphological specialisation in morphology, and it is through this ecologically driven filter that genotypic evolution takes place (Fig. 11). Given that the role of the directive genotype is efficiently to generate stable morphology, the modular structure of morphology is likely to have important consequences for genotypic evolution: (1) large bodies can be efficiently assembled as packets of similar units (i.e. segments or other repeated units) that can be individually specialised. Although a developmental system could use a different set of genes to build each unit, it would be more efficient to lay out the common morphological elements first and the differentiated ones later in a hierarchical developmental system. (2) The evolution of a primitive system with common repeated elements constructed by low-level genetic packages would generate evolutionary pressure for the common elements to be abstracted upwards into higher-level developmental packages. This would not only be more efficient,

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but would also be more stable – the result of canalising selection. One way in which this stability could be achieved would be by increasing the lethal effect of mutations involving such systems (because they would have global rather than local effects). (3) Morphology built by low-level genetic packages, perhaps within the morphogenotype, would act as a template for the construction of higher-level developmental systems. Specialisation in morphology will lead to special genetic packages to develop it. The constant optimising pressure of selection will thus tend to result in a correlation between modular phenotype and a modular developmental genotype. (4) The order of events in development will reflect degree of commonality of the structures being specified rather than the evolutionary order of acquisition of the structures, so that in this view, early stages of development would be open to great modification. (5) Only complex morphologies will favour complex high-level genotypic structures. For example, in an organism that has undergone complex low-level evolution, a duplication of the genome might be selectively favoured because only in such a system will the new genes thus generated be able to find new functions, even if they are initially retained in a functionally degenerate duplication. (6) The genetic origin of a structure is likely to be quite different from the genetic package that is later co-opted to stabilise it and build it more efficiently.

These suggestions grow out of the general functional morphological bias of the treatment of the origin of novelty herein. Nevertheless they stand in need of exploration and testing, and it must be stressed that the interaction between this 'modularising' pressure and that from the developmental process is not yet understood: indeed, no accepted theory for the origin of modularity currently exists.

IV. CONCLUSIONS

Biological systems may be considered to consist of ecological networks that provide the basic selective pressures acting on organisms, and genetic networks that build and maintain organisms. Much work has been carried out on understanding these two highly complex systems and how they might evolve (see Valentine, 2003). However, the neglected middle term is the structure of the organism itself; it is a complex network, and may have an essential mediating role between the two other systems. Only when ecological pressures are directed at vulnerable points in the organism system can change occur; and only then can ecological pressure affect the genetic network.

This latter interaction is complicated by the generative role the genome has for morphology, and there is a continuing need to understand the ways in which allowed morphological change is generated. One way is by microchanges in simple characters; another is by overall system change, for example by heterochronic change. The interaction of these two mechanisms with the models outlined herein may be fruitful to explore.

(2) Although morphological evolution has not always been critically analysed, (but see e.g. Lauder, 1981, Dullemeijer, 1981, Roth & Wake, 1989; and Galis, 1996 for exceptions), the examples given herein attempt to demonstrate that it can be categorised according to the less-constrained components of each system (functional asymmetries and redundancies) and how they are interconverted, freeing more constrained components. Further critical analysis of phenotypic systems in this light is required to understand patterns of phenotypic change; more categories of change may perhaps come to light as a result. Although the conventional, population-genetics view of evolution of complex structures is that morphology and its development are simultaneously constructed by the same selective pressures, I have tried to show above why this is unlikely for general systems. In particular, the functional thresholds may differ for different components of a system (indeed, that is partly how they are recognised as different components); secondly, genotypic and phenotypic short-term evolvability may not be appropriately geared.

(3) Such a view has certain implications for the origins of novelty. Clearly, it is unlikely that it is a 'gene-led' process where the genes in question are high-level developmental elements. Even in cases where the genetic basis for novelty is known, such as butterfly wing spots (Brakefield *et al.*, 1996), questions remain such as: why do they form where they are, and how did the interaction between the 'old' use of the patterning genes and the genes they control in their new role come about? Although it is sometimes assumed that mere ectopic expression of the old gene patterning package is enough to generate novelties, this seems hopelessly inadequate as an explanation. From the above analysis, a more likely scenario is that the directive genotype – the most constrained of the organism system – would be the last element to evolve, and would in general require the evolution of a system which it could appropriately pattern within the morphogenotype. This view of novelty allows but does not demand a highly nongenocentric view, and there seems to be no reason why the initial impetus for innovation could not be non-genetic, for example, by environmental perturbation of a developmental programme, outside interference by a pathogen (Budd, 2000) or other ecological cause. If such a cause was favoured, then, in line with structuralist and genetic assimilation ideas, such structures would be later infiltrated by genetic control networks and hierarchies (Seilacher, 1991; Müller & Newman, 1999, 2003). One compelling route for this might be developmental induction. While a particular tissue might not have an autonomous developmental programme, it can be induced to develop in a particular way by the proximity of another structure. Once this occurred, the nature and strength of the inductive process could be altered by selective pressure. In a classic genetic (e.g. Pigliucci, 2001) changeover of genetic control, autonomy of the new system could then be established without the usual developmental perturbation (in this instance, provided by another developing system of the organism). Indeed, instances of this change of control are reliably documented, for example in eye development (see discussion in Hall, 1998, p. 182), and hypothesised for the evolution of arthropod segmentation (Budd, 2001*b*). Inductive processes could in this manner establish developmental bridgeheads in other or newly evolving systems before full-blown genetic invasion took place. Finally, the

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developing field of epigenetics in the sense of imprinting (Gorelick & Betram, 2003) may also offer ways in which non-causal genetic information may influence gene expression and morphology.

While this is a broadly structuralist perspective, and thus is similar to the perspective of e.g. Müller & Newman (1999, 2003), it differs considerably in that the entire process is under selective control. There is no need to require an entire "character" (see Wagner, 2001) to emerge before genes become involved in its causal mechanism. Rather, as small components of morphology of a complex character are gradually assembled – through the routes outlined above – piecemeal co-option of genetic control mechanisms could also take place, albeit with a slight time-lag. As small-scale induced and ecophenotypic characters shift, they may themselves create pressure points in functionally integrated morphological systems, by introducing functional asymmetry, preadaptation and redundancies. The small amount of flexibility in the entire system that is an organism will thus slowly move, constantly under evolutionary pressure, while surprising novelties and deeply entrenched characters appear and dissolve.

(5) The study of functional morphology must thus stand at the heart of any attempt to understand the evolution of an organism as a whole, and genomic changes are as constrained by it as those of morphology itself. This insight supports a broadly structuralist account of evolution – phenotypic structure has primacy over the genome – but it does not support the more extreme versions which suggest that even compound characters could emerge through such processes. I have attempted to sketch a moderately comprehensive picture of how compound characters emerge and evolve. This suggests in principle how structuralist considerations can be worked into a population genetics and microevolutionary account of the changes involved in the origin of compound characters, even if population genetics itself does not reveal the ways in which these small-scale changes must be woven together to form the fabric of macroevolutionary change. The origins of innnovations may lie in small-scale events that may be genetic, but could also be environmentally or pathogenetically induced. In particular, developmental induction might be one way in which structures form that are later stabilised and canalised by genetic causal structures. The relationship between how morphology can evolve, and recent developments in the understanding of canalisation and assimilation (e.g. Stearns, 2002) may be a critical component of a holistic view of evolution.

V. ACKNOWLEDGEMENTS

This work has evolved over several years despite many constraints and has involved conversations with many colleagues, many of whom have made valuable, robust and critical suggestions. In particular, I would like to thank Olle Håstad, Lennart Olsson, Frietson Galis, Günter Wagner, Wendy Olson, Michael Akam, David Stern and Kurt Schwenk; their agreement with the content is not necessarily complete. Alan Love kindly corrected the manuscript. I note, with regret, the death of Rupert Riedl during the preparation of this work. This work was funded by the Swedish Research Council (VR) and the Swedish Royal Academy of Sciences (KVA), both of whom are gratefully acknowledged.

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https://doi.org/10.1017/S1464793106007135 Published online by Cambridge University Press